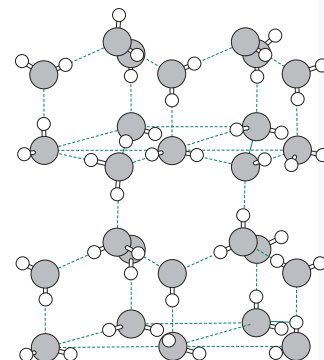


Simple Bonding Theory



We now turn from the use of quantum mechanics and its description of the atom to an elementary description of molecules. Although most of our discussion of chemical bonding uses the molecular orbital approach, less rigorous methods that provide approximate pictures of the shapes and polarities of molecules are also useful. This chapter provides an overview of Lewis dot structures, valence shell electron-pair repulsion (VSEPR), and related topics. Molecular orbital descriptions of some of the same molecules are presented in Chapter 5 and later chapters; the ideas of this chapter provide a starting point for that more modern treatment.

Ultimately, any description of bonding must be consistent with experimental data on bond lengths, bond angles, and bond strengths. Angles and distances are most frequently determined by diffraction (X-ray crystallography, electron diffraction, neutron diffraction) or spectroscopic (microwave, infrared) methods. For many molecules, there is general agreement on the nature of the bonding, although there are alternative ways to describe it. For others, there is considerable difference of opinion on the best way to describe the bonding. In this chapter and in Chapter 5, we describe some useful qualitative approaches, including some of the opposing views.

3.1 Lewis Electron-Dot Diagrams

Lewis electron-dot diagrams, although oversimplified, provide a good starting point for analyzing the bonding in molecules. Credit for their initial use goes to G. N. Lewis,¹ an American chemist who contributed much to the understanding of thermodynamics and chemical bonding in the early twentieth century. In Lewis diagrams, bonds between two atoms exist when they share one or more pairs of electrons. In addition, some molecules have nonbonding pairs, also called *lone pairs*, of electrons on atoms. These electrons contribute to the shape and reactivity of the molecule but do not directly bond the atoms together. Most Lewis structures are based on the concept that eight **valence electrons**, corresponding to *s* and *p* electrons outside the noble gas core, form a particularly stable arrangement, as in the noble gases with $s^2 p^6$ configurations. An exception is hydrogen, which is stable with two valence electrons. Also, some molecules require more than eight electrons around a given central atom, and some molecules require fewer than eight electrons.

Simple molecules such as water follow the **octet rule**, in which eight electrons surround the central atom. Each hydrogen atom shares two electrons with the oxygen, forming the familiar structure with two bonds; the O atom accommodates two bonding pairs and two lone pairs:*



*We will see in Chapter 5 that the treatment of water via molecular orbital theory results in an electronic structure in which each of these electron pairs has a unique energy. This model is supported by spectroscopic evidence, and indicates one limitation of the Lewis model.

Shared electrons are considered to contribute to the electronic requirements of both atoms involved; thus, the electron pairs shared by H and O in the water molecule are counted toward both the 8-electron requirement of oxygen and the 2-electron requirement of hydrogen.

The Lewis model defines double bonds as containing four electrons and triple bonds as containing six electrons:



3.1.1 Resonance

In many Lewis structures, the choice of which atoms are connected by multiple bonds is arbitrary. When alternate locations for single bonds and multiple bonds are possible that all afford valid Lewis structures, a structure demonstrating each option should be drawn. For example, three drawings (resonance structures) of CO_3^{2-} are needed (Figure 3.1) to show the double bond in each of the three possible C—O positions. In fact, experimental evidence shows that all three C—O bonds are equivalent, with bond lengths (129 pm) between typical C—O double-bond and single-bond distances (116 pm and 143 pm, respectively). All three drawings are necessary to describe the structure, with each drawing contributing equally to describe the bonding in the actual ion. This is called **resonance**; there is more than one possible way in which the valence electrons can be placed in a Lewis structure. Note that in resonance structures, such as those shown for CO_3^{2-} in Figure 3.1, the electrons are arranged differently, but the nuclei remain in fixed positions.

The species CO_3^{2-} and NO_3^- have the same number of electrons (*i.e.*, they are **isoelectronic**) and use the same orbitals for bonding. Their Lewis diagrams are identical except for the identity and formal charge (Section 3.1.3) of the central atom.

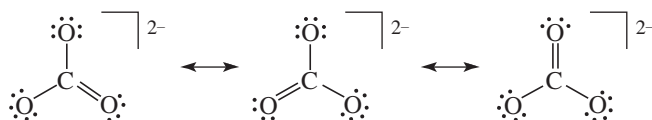
When a molecule has several resonance structures, its overall electronic energy is lowered, making it more stable. Just as the energy levels of a particle in a box are lowered by making the box larger, the electronic energy levels of the bonding electrons are lowered when the electrons can occupy a larger space. The molecular orbital description of this effect is presented in Chapter 5.

3.1.2 Higher Electron Counts

When it is impossible to draw a structure consistent with the octet rule because additional valence electrons remain to be assigned after the octet rule is satisfied on all atoms, it is necessary to increase the number of electrons around the central atom. An option limited to elements of the third and higher periods is to use *d* orbitals for this expansion, although theoretical work suggests that expansion beyond the *s* and *p* orbitals is unnecessary for most main group molecules.² In most cases, two or four added electrons will complete the bonding, but more can be added if necessary. For example, 10 electrons are required around chlorine in ClF_3 and 12 around sulfur in SF_6 (Figure 3.2). The increased number of electrons is often described as an *expanded shell* or an *expanded electron count*. The term **hypervalent** is used to describe central atoms that have electron counts greater than the atom's usual requirement.

There are examples with even more electrons around the central atom, such as IF_7 (14 electrons), $[\text{TaF}_8]^{3-}$ (16 electrons), and $[\text{XeF}_8]^{2-}$ (18 electrons). There are rarely more than 18 electrons (2 for *s*, 6 for *p*, and 10 for *d* orbitals) around a single atom in the top half of the periodic table, and crowding of the outer atoms usually keeps the number below this, even for much heavier atoms that have *f* orbitals energetically available.

FIGURE 3.1 Lewis Diagrams for CO_3^{2-} .



3.1.3 Formal Charge

Formal charge is the apparent electronic charge of each atom in a molecule, based on the electron-dot structure. Formal charges help assess resonance structures and molecular topology, and they are presented here as a simplified method of describing structures, just as the Bohr model is a simple method of describing electronic configurations in atoms. Both of these methods have limitations, and other approaches are more accurate, but they can be useful as long as their imperfections are kept in mind.

Formal charges can help in eliminating resonance structures expected to contribute very little to the electronic ground state of the molecule, and, in some cases, suggesting multiple bonds beyond those required by the octet rule. It is essential, however, to remember that formal charge is only a tool for assessing Lewis structures, not a measure of any actual charge on the atoms. The number of valence electrons available in a free atom of an element minus the total for that atom in the molecule—determined by counting lone pairs as two electrons and bonding pairs as one electron assigned to each atom—is the formal charge on the atom:

$$\text{Formal charge} = \left(\begin{array}{c} \text{number of valence} \\ \text{electrons in a free} \\ \text{atom of the element} \end{array} \right) - \left(\begin{array}{c} \text{number of unshared} \\ \text{electrons on the atom} \end{array} \right) - \left(\begin{array}{c} \text{number of bonds} \\ \text{to the atom} \end{array} \right)$$

In addition,

$$\text{Charge on molecule or ion} = \text{sum of formal charges}$$

Resonance structures that contribute more to the electronic ground state of the species generally (a) have smaller magnitudes of formal charges, (b) place negative formal charges on more electronegative elements (in the upper right-hand part of the periodic table), and (c) have smaller separation of charges. Three examples— SCN^- , OCN^- , and CNO^- —will illustrate the use of formal charges in describing electronic structures.

EXAMPLE 3.1

SCN^-

In the thiocyanate ion, SCN^- , three resonance structures are consistent with the electron-dot method, as shown in **Figure 3.3**. Structure A has only one negative formal charge on the nitrogen atom, the most electronegative atom in the ion. Structure B has a single negative charge on the S, which is less electronegative than N. Structure C has charges of $2-$ on N and $1+$ on S, consistent with the relative electronegativities of these atoms but also has a large magnitude $2-$ charge and greater charge separation than the other structures. Therefore these structures lead to the prediction that structure A contributes the most to the electronic ground state of SCN^- , structure B contributes an intermediate amount, and any contribution from C is minor in describing the electronic ground state of SCN^- .

The bond lengths in **Table 3.1** are somewhat consistent with this conclusion, with SCN^- bond lengths between those of structures A and B. Protonation of the ion forms HNCS, consistent with a negative charge on N in SCN^- . The bond lengths in HNCS are close to those of double bonds, consistent with the structure $\text{H}-\text{N}=\text{C}=\text{S}$.

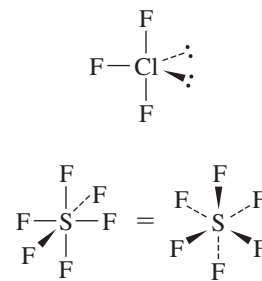
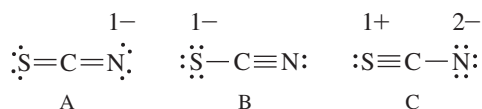


FIGURE 3.2 Structures of ClF_3 and SF_6 .

FIGURE 3.3 Resonance Structures of Thiocyanate, SCN^- .

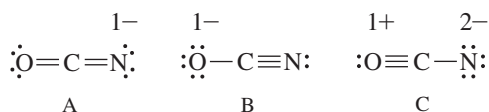
TABLE 3.1 Table of S—C and C—N Bond Lengths (pm)

	S—C	C—N
SCN [−] (in NaSCN)	165	118
HNCS	156	122
Single bond	181	147
Double bond	155	128 (approximate)
Triple bond		116

Data from A. F. Wells, *Structural Inorganic Chemistry*, 5th ed., Oxford University Press, New York, 1984, pp. 807, 926, 934–936.

EXAMPLE 3.2**OCN[−]**

The isoelectronic cyanate ion, OCN[−] (Figure 3.4), has the same possibilities, but the larger electronegativity of O is expected to make structure B contribute more to the electronic ground state in cyanate relative the contribution of B in thiocyanate. The protonation of cyanate results in two isomers, 97% HNCO and 3% HOCN, consistent with a major contribution of structure A and a small, but significant, contribution from B. The bond lengths in OCN[−] and HNCO in Table 3.2 are reasonably consistent with this analysis. Formal charge arguments provide a good starting point to assess Lewis structures, and reactivity patterns are also useful to gain experimental insight about electron distributions.

FIGURE 3.4 Resonance Structures of Cyanate, OCN[−].**TABLE 3.2** Table of O—C and C—N Bond Lengths (pm)

	O—C	C—N
OCN [−]	126	117
HNCO	118	120
Single bond	143	147
Double bond	116 (CO ₂)	128 (approximate)
Triple bond	113 (CO)	116

Data from A. F. Wells, *Structural Inorganic Chemistry*, 5th ed., Oxford University Press, New York, 1984, pp. 807, 926, 933–934; S. E. Bradforth, E. H. Kim, E. W. Arnold, D. M. Neumark, *J. Chem. Phys.*, **1993**, 98, 800.

EXAMPLE 3.3**CNO[−]**

The isomeric fulminate ion, CNO[−] (Figure 3.5), can be drawn with three similar structures, but the resulting formal charges have larger magnitudes than in OCN[−]. Because the order of electronegativities is C < N < O, none of these are ideal structures, and it is not surprising that this ion is unstable. The only common fulminate salts are of mercury and silver; both are explosive. Fulminic acid is linear HCNO in the vapor phase, consistent with the greatest contribution from structure C; coordination complexes of CNO[−] with transition-metal ions are known with MCNO structures.³

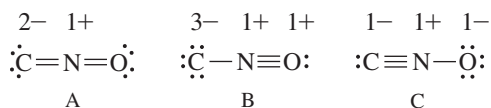


FIGURE 3.5 Resonance Structures of Fulminate, CNO^- .

EXERCISE 3.1 Use electron-dot diagrams and formal charges to predict the bond order for each bond in POF_3 , SOF_4 , and SO_3F^- .

Some molecules have satisfactory electron-dot structures with octets but have more reasonable formal charge distributions in their structures with expanded electron counts. In each of the cases in **Figure 3.6**, the actual molecules and ions are consistent with electron counts greater than 8 on the central atom and with a large contribution from the resonance structure that uses multiple bonds to minimize formal charges. The multiple bonds may also influence the shapes of the molecules.

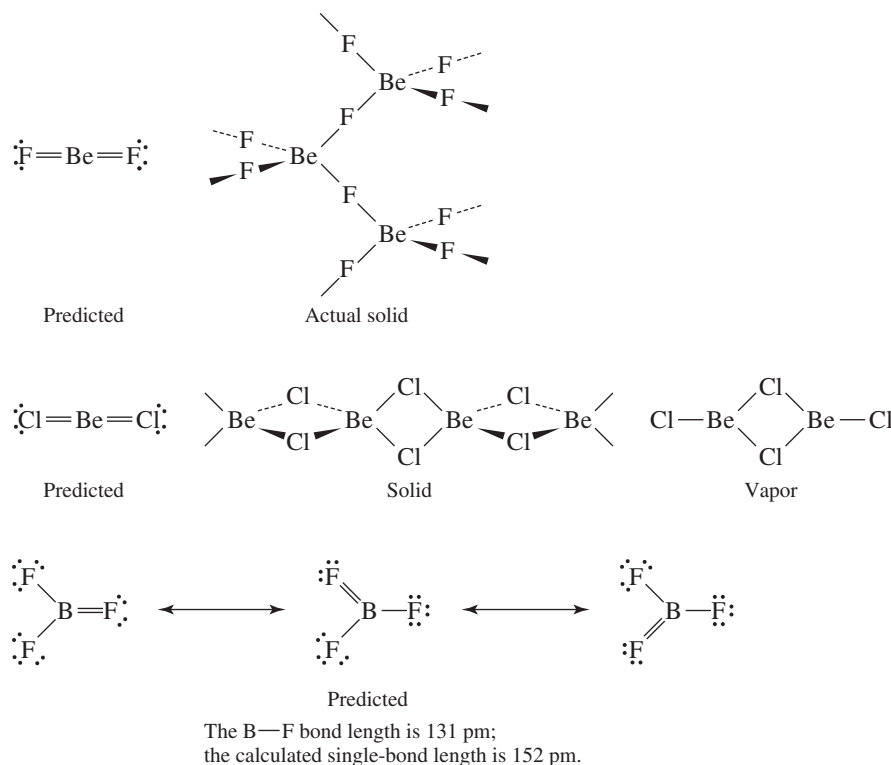
3.1.4 Multiple Bonds in Be and B Compounds

A few molecules—such as BeF_2 , BeCl_2 , and BF_3 —seem to require multiple bonds to satisfy the octet rule for Be and B, even though multiple bonds for F and Cl are not generally expected on the basis of the high electronegativities of these halogens. Structures minimizing formal charges for these molecules have only four electrons in the valence shell of Be and six electrons in the valence shell of B, in both cases fewer than the usual octet. The alternative, requiring eight electrons on the central atom, predicts multiple bonds, with BeF_2 analogous to CO_2 and BF_3 analogous to SO_3 (**Figure 3.7**). These structures, however, result in nonideal formal charges (2− on Be and 1+ on F in BeF_2 , and 1− on B and 1+ on the double-bonded F in BF_3) on the basis of the usual rules.

Octet				Expanded			
Molecule		Atom	Formal Charge		Atom	Formal Charge	Expanded to:
SNF ₃	<pre> :N: :F--S--F: :F:</pre>	S N	2+ 2-	<pre> :N: :F--S--F: :F:</pre>	S N	0 0	12
SO ₂ Cl ₂	<pre> :O: :Cl--S--O: :Cl:</pre>	S O	2+ 1-	<pre> :O: :Cl--S--O: :Cl:</pre>	S O	0 0	12
XeO ₃	<pre> :O: :O--Xe--O:</pre>	Xe O	3+ 1-	<pre> :O: :O=Xe=O:</pre>	Xe O	0 0	14
SO ₃ ²⁻	<pre> :O: :O--S--O:</pre>	S O	1+ 1-	<pre> :O: :O--S--O:</pre>	S O	0 0,1-	10

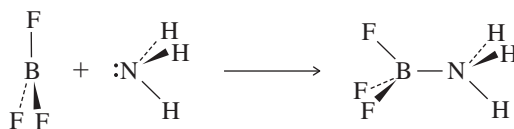
FIGURE 3.6 Formal Charge and Expanded Electron Counts on Central Atom.

FIGURE 3.7 Structures of BeF_2 , BeCl_2 , and BF_3 . (Data from A. F. Wells, *Structural Inorganic Chemistry*, 5th ed., Oxford University Press, Oxford, England, 1984, pp. 412, 1047.)



In solid BeF_2 , a complex network is formed with a Be atom coordination number of 4 (see Figure 3.7). BeCl_2 dimerizes to a 3-coordinate structure in the vapor phase, but the linear monomer is formed at high temperatures. This monomeric structure is unstable due to the electronic deficiency at Be; in the dimer and the network formed in the solid-state, the halogen atoms share lone pairs with the Be atom in an attempt to fill beryllium's valence shell. The monomer is still frequently drawn as a singly bonded structure, with only four electrons around the beryllium and the ability to accept lone pairs of other molecules to relieve its electronic deficiency (Lewis acid behavior, discussed in Chapter 6).

Bond lengths in all the boron trihalides are shorter than expected for single bonds, so the partial double-bond character predicted seems reasonable despite the nonideal formal charges of these resonance forms. While a small amount of double bonding is possible in these molecules, the strong polarity of the B—halogen bonds and the ligand close-packing (LCP) model (Section 3.2.4) have been used to account for the short bonds without the need to invoke multiple bonding. The boron trihalides combine readily with other molecules that can contribute a lone pair of electrons (Lewis bases), forming a roughly tetrahedral structure with four bonds:



Because of this tendency, boron trihalides are frequently drawn with only six electrons around the boron.

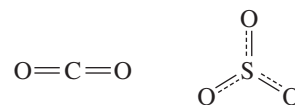
Other boron compounds that cannot be adequately described via simple electron-dot structures include hydrides such as B_2H_6 , and many more complex molecules. Their structures are discussed in Chapters 8 and 15.

3.2 Valence Shell Electron-Pair Repulsion

Valence shell electron-pair repulsion (VSEPR) is an approach that provides a method for predicting the shape of molecules based on the electron-pair electrostatic repulsion described by Sidgwick and Powell⁴ in 1940 and further developed by Gillespie and Nyholm⁵ in 1957 and in the succeeding decades. Despite this method's simple approach, based on Lewis electron-dot structures, the VSEPR method in most cases predicts shapes that compare favorably with those determined experimentally. However, this approach at best provides approximate shapes for molecules. The most common method of determining the actual structures is X-ray diffraction, although electron diffraction, neutron diffraction, and many spectroscopic methods are also used.⁶ In Chapter 5, we will provide molecular orbital approaches to describe bonding in simple molecules.

The basis of the VSEPR approach is that electrons repel each other because they are negatively charged. Quantum mechanical rules dictate that electrons can be accommodated in the same region of space as bonding pairs or lone pairs, but each pair repels all other pairs. According to the VSEPR model, therefore, molecules adopt geometries such that valence electron pairs position themselves as far from each other as possible to minimize electron–electron repulsions. A molecule can be described by the generic formula AX_mE_n , where A is the central atom, X stands for any atom or group of atoms surrounding the central atom, and E represents a lone pair of electrons. The **steric number*** ($SN = m + n$) is the total number of positions occupied by atoms or lone pairs around a central atom; lone pairs and bonding pairs both influence the molecular shape.

Carbon dioxide is a molecule with two atoms attached ($SN = 2$) to the central atom via double bonds. The electrons in each double bond must be between C and O, and the repulsion between these electron groups forces a linear structure on the molecule. Sulfur trioxide has three atoms bound to the sulfur ($SN = 3$), with equivalent partial double-bond character between sulfur and each oxygen, a conclusion rendered by analysis of its resonance forms. The best positions for the oxygens to minimize electron–electron repulsions in this molecule are at the corners of an equilateral triangle, with $O-S-O$ bond angles of 120° . The multiple bonding does not affect the geometry, because all three bonds are equivalent in terms of bond order.



The same pattern of finding the Lewis structure and then matching it to a geometry that minimizes the repulsive energy of bonding electrons is followed through steric numbers 4, 5, 6, 7, and 8 where the outer atoms are identical in each molecule, as shown in **Figure 3.8**.

Bond angles and distances are uniform in each of these structures with two, three, four, and six electron pairs. Neither the corresponding 5- nor 7-coordinate structures can have uniform angles and distances, because there are no regular polyhedra with these numbers of vertices. The 5-coordinate molecules have a trigonal bipyramidal structure, with a central triangular plane of three positions plus two other positions above and below the center of the plane. The 7-coordinate molecules have a pentagonal bipyramidal structure, with a pentagonal plane of five positions and positions above and below the center of the plane. The regular square antiprism structure ($SN = 8$) is like a cube that has had the top face twisted 45° into the antiprism arrangement, as shown in **Figure 3.9**. It has three different bond angles for adjacent fluorines. $[TaF_8]^{3-}$ has square antiprismatic geometry but is distorted from this ideal in the solid.⁷

*The steric number is also called the number of electron pair domains.

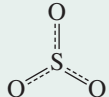
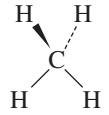
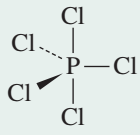
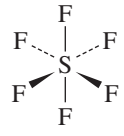
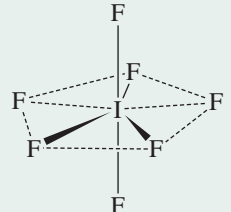
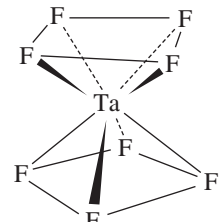
Steric Number	Geometry	Examples	Calculated Bond Angles	
2	Linear	CO_2	180°	$\text{O}=\text{C}=\text{O}$
3	Trigonal (triangular)	SO_3	120°	
4	Tetrahedral	CH_4	109.5°	
5	Trigonal bipyramidal	PCl_5	$120^\circ, 90^\circ$	
6	Octahedral	SF_6	90°	
7	Pentagonal bipyramidal	IF_7	$72^\circ, 90^\circ$	
8	Square antiprismatic	$[\text{TaF}_8]^{3-}$	$70.5^\circ, 99.6^\circ, 109.5^\circ$	

FIGURE 3.8 VSEPR Predictions.

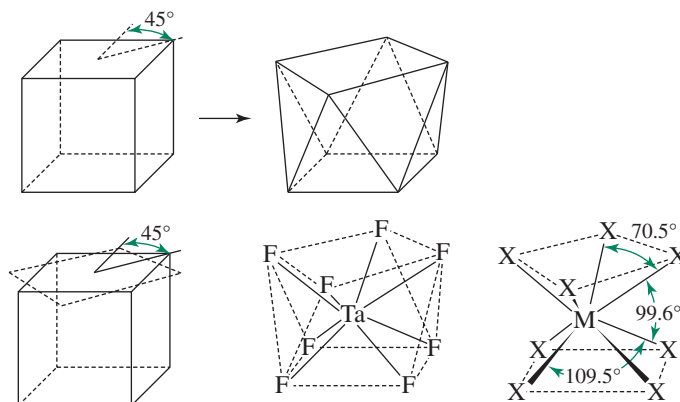


FIGURE 3.9 Conversion of a Cube into a Square Antiprism.

3.2.1 Lone-Pair Repulsion

Bonding models are useful only if their explanations are consistent with experimental data. New theories are continually being suggested and tested. Because we are working with such a wide variety of atoms and molecular structures, a single approach will unlikely work for all of them. Although the fundamental ideas of atomic and molecular structures are relatively simple, their application to complex molecules is not. To a first approximation, lone pairs, single bonds, double bonds, and triple bonds can all be treated similarly when predicting molecular shapes. However, better predictions of overall shapes can be made by considering some important differences between lone pairs and bonding pairs. These methods are sufficient to show the trends and explain the bonding, as in rationalizing why the $\text{H}-\text{N}-\text{H}$ angle in ammonia is smaller than the tetrahedral angle in methane and larger than the $\text{H}-\text{O}-\text{H}$ angle in water.

As a general guideline, the VSEPR model predicts that electron-pair repulsions involving lone pairs (*lp*) are stronger than those involving bonding pairs (*bp*) in the order

$$lp-lp \text{ repulsions} > lp-bp \text{ repulsions} > bp-bp \text{ repulsions}$$

Steric Number = 4

The isoelectronic molecules CH_4 , NH_3 , and H_2O (Figure 3.10) illustrate the effect of lone pairs on molecular shape. Methane has four identical bonds between carbon and each of the hydrogens. When the four pairs of electrons are arranged as far from each other as possible, the result is the familiar tetrahedral shape. The tetrahedron, with all $\text{H}-\text{C}-\text{H}$ angles measuring 109.5° , has four identical bonds.

Ammonia also has four pairs of electrons around the central atom, but three are bonding pairs between N and H, and the fourth is a lone pair on the nitrogen. The nuclei form a trigonal pyramid with the three bonding pairs; the lone pair occupies the fourth region in space resulting in a tetrahedral arrangement of the four electron groups. Because each of the three bonding pairs is attracted by two positively charged nuclei (H and N), these pairs are largely confined to the regions between the H and N atoms. The lone pair, on the other hand, is attracted solely by the nitrogen nucleus; it has no second nucleus to confine it to a small region of space. Consequently, the lone pair tends to spread out and to occupy more space around the nitrogen than the bonding pairs. As a result, the $\text{H}-\text{N}-\text{H}$ angles are 106.6° , nearly 3° smaller than the angles in methane.

The same principles apply to the water molecule, in which two lone pairs and two bonding pairs repel each other. Again, the electron pairs adopt a nearly tetrahedral arrangement, with the atoms arranged in a V shape. The angle of largest repulsion, between the two lone pairs, cannot be measured. However, the lone pair–bonding pair (*lp–bp*) repulsion is greater than the bonding pair–bonding pair (*bp–bp*) repulsion; as a result, the $\text{H}-\text{O}-\text{H}$ bond angle is only 104.5° , another 2.1° decrease from the ammonia angles. The net result is that we can predict approximate molecular shapes by assigning more space to lone electron pairs; lone pairs are able to spread out and occupy more space since they are attracted to one nucleus rather than two.

Steric Number = 5

For the trigonal bipyramidal geometry, there are two unique locations for electron pairs, axial and equatorial. If there is a single lone pair, for example in SF_4 , the lone pair occupies an equatorial position. This position provides the lone pair with the most space and minimizes the interactions between the lone pair and bonding pairs. If the lone pair were axial, it would have three 90° interactions with bonding pairs; in an equatorial position, it has only two such interactions, as shown in Figure 3.11. The actual structure is distorted by the lone pair as it spreads out in space and effectively squeezes the rest of the molecule together.

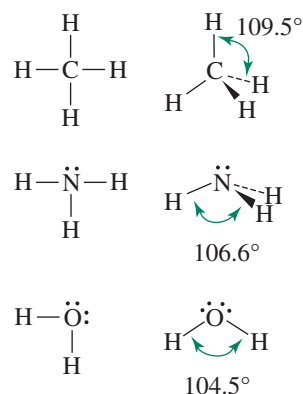


FIGURE 3.10 Shapes of Methane, Ammonia, and Water.

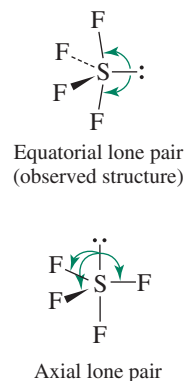


FIGURE 3.11 Possible Structures of SF_4 .

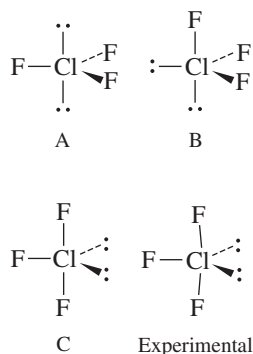


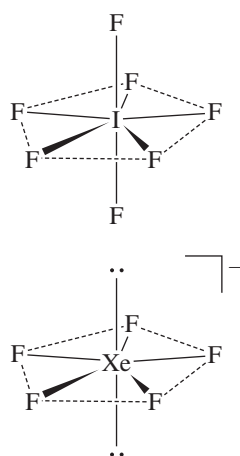
FIGURE 3.12 Possible Structures of ClF_3 .

ClF_3 provides a second example of the influence of lone pairs in molecules having a steric number of 5. There are three possible structures for ClF_3 , as shown in **Figure 3.12**.

In determining the feasibility of different structures, lone pair–lone pair interactions should be considered first, followed by lone pair–bonding pair interactions. These interactions at angles of 90° or less are generally considered destabilizing; larger angles generally render structures more feasible. For example, in ClF_3 , structure B can be eliminated quickly because of the 90° $lp-lp$ angle. The $lp-lp$ angles are large for A and C, so the choice must come from the $lp-bp$ and $bp-bp$ angles. Because the $lp-bp$ angles are more important, C, which has only four 90° $lp-bp$ interactions, is favored over A, which has six such interactions. Experiments have confirmed that the structure is based on C, with slight distortions due to the lone pairs. The lone pair–bonding pair repulsion causes the $lp-bp$ angles to be larger than 90° and the $bp-bp$ angles to be less than 90° (actually, 87.5°). The $\text{Cl}-\text{F}$ bond distances show the repulsive effects as well, with the axial fluorines (approximately 90° $lp-bp$ angles) at 169.8 pm and the equatorial fluorine (in the plane with two lone pairs) at 159.8 pm.⁸ Angles involving lone pairs cannot be determined experimentally.

Interaction	Angles in Possible Structures			Experimental
	A	B	C	
$lp-lp$	180°	90°	120°	Cannot be determined
$lp-bp$	6 at 90°	3 at 90°	4 at 90°	Cannot be determined
		2 at 120°	2 at 120°	
$bp-bp$	3 at 120°	2 at 90°	2 at 90°	2 at 87.5°
		1 at 120°		Axial $\text{Cl}-\text{F}$ 169.8 pm Equatorial $\text{Cl}-\text{F}$ 159.8 pm

Additional examples of structures with lone pairs are illustrated in **Figure 3.13**. The structures based on a trigonal bipyramidal arrangement of electron pairs around a central atom always place any lone pairs in the equatorial plane, as in SF_4 , BrF_3 , and XeF_2 . The resulting shapes minimize both lone pair–lone pair and lone pair–bonding pair repulsions. The shapes are called *seesaw* (SF_4), *distorted T* (BrF_3), and *linear* (XeF_2).



Steric Numbers = 6 and 7

In octahedral structures, all six positions are equivalent. When a single lone pair is present, it typically repels adjacent bonding pairs, reducing bond angles accordingly, as for IF_5 in **Figure 3.13**. In octahedron-based structures with two lone pairs, lone pair–lone pair repulsion is minimized if these pairs are *trans*, and this is the shape that is adopted. Square planar XeF_4 , also shown in **Figure 3.13**, is an example. Recently XeF_3^- , which would be expected to have a steric number of 6 and three lone pairs, has been reported in the gas phase, but attempts to prepare salts of this ion have been unsuccessful.⁹

The shape that minimizes electron-pair repulsions for a steric number of 7 is the pentagonal bipyramid, shown in **Figure 3.8**. IF_7 (in the margin) and TeF_7^{2-} exhibit this shape, with both axial and equatorial fluorines. If a single lone pair is present, in some cases the lone pair causes distortion. The nature of this distortion is not always easy to ascertain; XeF_6 is a classic example.¹⁰ In other cases the structure is octahedral (see Problem 3.26) with the lone pair not stereochemically active.* Two lone pairs minimize their repulsions by adopting axial (*trans*) positions, with the atoms all in the equatorial plane. Two known examples are XeF_5^- (in the margin) and IF_5^{2-} .

*A lone pair that appears in the Lewis-dot structure but has no apparent effect on the molecular geometry is classified as not stereochemically active. The VSEPR model assumes that all lone pairs are stereochemically active and therefore do affect the molecular geometry.